REM tutorial

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Contents

Part I

Introduction to relational event models

1 Core REMs

Relational event sequence: realization of a **marked point process**.

$$E = \{e_i = (s_i, r_i, t_i), i = 1, \ldots, n\}, \tag{1}$$

where:

- s_i: **sender** of the i-th relational event;
- r_i: **receiver** of the i-th relational event;
- t_i: **time** of the i-th relational event;
- n: **number of events**.
- i: **index of a generic event**.

Longitudinal network / multivariate counting process [Perry and Wolfe, 2013]:

$$\{N_{sr}(t)\}_{s \in V^S,\, r \in V^R,\, t \geq 0}, \tag{2}$$

where:

- $N_{sr}(t)$: number of interactions from s to r until time t.
- V^S: **set of senders** in the system;
- V^R: **set of receivers** in the system;

If senders/receivers belong to the same group, then $V^S = V^R = V$ (**vertex set**), with $p = |V|$.

Doob-Meyer decomposition theorem:

$$N_{sr}(t) = \Lambda_{sr}(t) + M_{sr}(t) = \int_0^t \lambda_{sr}(u)\mathrm{d}u + M_{sr}(t)$$

where:

- $\Lambda_{sr}(t)$: **cumulative intensity process** (structural part);
- $M_{sr}(t)$: **Martingale residual process** (noisy part);
- $\lambda_{sr}(t)$: **hazard** (also: **rate**, **intensity**): instantaneous rate at which (s, r) occurs at t.

Relational event model (REM) [Bianchi et al., 2024]:

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{f(\boldsymbol{x}_{sr}(t), t)\} \tag{3}$$

where:

- $W_{sr}(t)$: **risk indicator** – it is equal to 1 if (s, r) is at risk at time t;
- $\lambda_0(t)$: **baseline hazard** (including global risk determinants);
- $f(\boldsymbol{x}_{sr}(t), t)$: **event log-rate contribution**. E.g. **Linear contribution** $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t)$
  - $f(\cdot, t)$: **contribution function** (also, **effect**);
  - $\boldsymbol{x}_{sr}(t)$: edge-specific **covariates**;
  - t: **calendar time**.

The dyads (s, r) satisfying $W_{sr}(t) = 1$ compose the **risk set** $\mathcal{R}_t$.

Gillespie algorithm for simulating REMs:

$$\Delta T_{sr}(t) \sim \text{Exponential}(\lambda_{sr}(t)) ,$$

$$T = \min_{(s,r)\in\mathcal{R}_t} \Delta T_{sr}(t) \sim \text{Exponential}\left(\sum_{(s,r)\in\mathcal{R}_t} \lambda_{sr}(t) \right) ,$$

$$D \sim \text{Multinomial}\big(\{p_{sr}(t)\}_{(s,r)\in\mathcal{R}_t}\big) ,$$

$$p_{sr}(t) = \frac{\lambda_{sr}(t)}{\sum_{(s,r)\in\mathcal{R}_t} \lambda_{sr}(t)}, \quad (s,r) \in \mathcal{R}_t .$$

where:

- $\Delta T_{sr}(t)$ is the **waiting time** (random variable) until an event involving sender s and receiver r occurs, evaluated at time t.
- T is the waiting time (random variable) until the next event, evaluated at time t.
- D (random variable) identifies the dyad (s, r) involved in the event occurring at time $t+T$.

Assumption: $\lambda_{sr}(t)$ does not change between t and $t + T$.

Exogenous & endogenous drivers: measured by means of **covariates** $x_{sr}(t)$. Covariates can be seen as aggregations/summaries of information derived from **attributes** $z_{sr}(t)$ of nodes and edges.

- **Exogenous covariates**: do not depend on the network of (previous) events.

  E.g. **Work experience dissimilarity**, obtained as the absolute value of the difference in **work experience** between the two nodes involved in the event.

  $$x_{sr}(t) = |WE_s - WE_r|$$

- **Endogenous covariates**: depend on the network of (previous) events and allow us to measure **relational mechanisms**.

  E.g. **Reciprocity**.

  - **Binary covariates**: check for the presence (in the previous event history) of **at least one event satisfying a condition** of interest.

    E.g. Is there **at least one reciprocal event** in the past?

    $$x_{sr}(t) = \mathbf{1}\big(\{\, (r, s, t^*) \mid t^* < t \,\} \neq \varnothing\big)$$

  - **Count covariates**: count the **number of events satisfying a condition** of interest.

    E.g. **How many reciprocal events** occurred in the past?

    $$x_{sr}(t) = \big|\{\, (r, s, t^*) \mid t^* < t \,\}\big|$$

  - **Time-aware covariates**: defined as a function of the **internal time** of the mechanism of interest (i.e., the difference between the current time and the time of the event satisfying the condition of interest in the previous event history).

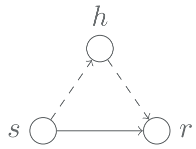
    E.g. **How recent is the reciprocal event** in the past?

    $$z_{sr}(t) = \begin{cases} t - \tau_{sr}, & \tau_{sr} = \text{time of the last event } (r, s), \\ \infty, & \text{otherwise,} \end{cases} \qquad x_{sr}(t) = \exp\big(-z_{sr}(t)\big)$$

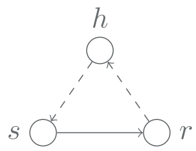
Relational mechanisms can be **dyadic** or **triadic**, depending on the number of nodes involved in the structure of interest. Some examples:

Repetition (dyadic):

Transitive closure (triadic):



Cyclic closure (triadic):



2 REM Inference

Assumption (this section only): **linear contribution function**, i.e., $f(\boldsymbol{x}_{sr}(t), t) = \boldsymbol{\beta} \cdot \boldsymbol{x}_{sr}(t)$.

The counting process (18) is adapted to the **filtration** $\mathbb{W} = \{\mathcal{W}_t\}_{t \in \mathbb{R}^+}$. At time t, $\mathcal{W}_{t^-}$ incorporates both exogenous information and information from events occurred at $t^* < t$.

Likelihood: constructed as joint probability of observing **data** (1) under **model** (19).

Preliminaries: Given a set of event times $t_1, t_2, \dots, t_n$, which are realizations of a point process:

- the associated counting process is a right-continuous function,
- that jumps by one at each event time and is constant otherwise,
- with conditional intensity function be $\lambda(t \mid \mathcal{W}_{t^-})$

the joint probability density of observing exactly n event times in the interval $(0, t_n)$ is

$$f(t_1, t_2, \dots, t_n) = \prod_{i=1}^{n} \lambda(t_i \mid \mathcal{W}_{t_i^-}) \exp\left\{ -\int_{t_{i-1}}^{t_i} \lambda(u \mid \mathcal{W}_{u^-}) \, \mathrm{d}u \right\}.$$

2.1 Full likelihood approaches

Full likelihood [Butts, 2008]:

$$\mathcal{L}(\boldsymbol{\beta}) = \prod_{i=1}^{n} p(t_i | \mathcal{W}_{t_i^-}; \boldsymbol{\beta}) \times p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \boldsymbol{\beta}) \tag{4}$$

$$= \prod_{i=1}^{n} \underbrace{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i^-}; \boldsymbol{\beta}) \exp\left[-\sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u | \mathcal{W}_u; \boldsymbol{\beta}) \mathrm{d}u \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \underbrace{\frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i^-}; \boldsymbol{\beta})}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i^-}; \boldsymbol{\beta})}}_{\text{Multinomial Probability}} \tag{5}$$

Piecewise Constant Rates (PCR) assumption [Stadtfeld et al., 2017]: event rate is assumed to be constant within inter-arrival times $\Rightarrow$ Exponential **waiting time** $\Delta T_i = T_i - T_{i-1}$ with Poisson number of events in ΔT_i.

$$\mathcal{L}(\boldsymbol{\beta}) \overset{\text{PCR}}{=} \prod_{i=1}^{n} \lambda_{s_i r_i}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta}) \prod_{(s,r)\in\mathcal{R}_{t_i}} \exp\left(-\lambda_{sr}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta})\,(t_i - t_{i-1}) \right) \tag{6}$$

Letting $\Delta N_{s_i r_i} = N_{s_i r_i}(t_i) - N_{s_i r_i}(t_{i-1})$,

$$\mathcal{L}(\boldsymbol{\beta}) \overset{\text{PCR}}{=} \prod_{i=1}^{n} \prod_{(s,r)\in\mathcal{R}_{t_i}} \left(\lambda_{sr}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta})\right)^{\Delta N_{s_i r_i}} \exp\left(-\lambda_{sr}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta})\,(t_i - t_{i-1}) \right) \tag{7}$$

is proportional to the likelihood of a **Poisson Regression**.

Full log-likelihood in practice (PCR)

Log-likelihood :

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \sum_{(s,r)\in\mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log\left(\lambda_{sr}(t_i;\boldsymbol{\beta})\right) - \lambda_{sr}(t_i;\boldsymbol{\beta})(t_i - t_{i-1})$$

Inference technique : **Poisson regression**;

$$\Delta N_{sr}(t_i)|\boldsymbol{x}_{sr}(t_i) \overset{\text{iid}}{\sim} \text{Poisson}\left(\mu_{sr}(t_i)\right), \qquad \log(\mu_{sr}(t_i)) = \boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

Data overview & R code :

```
> head(poisson_dat, 10)
   sender receiver dist_min  dist   PT_dist random_intercept_s info event  start        stop log_dist     logtm
1   33202    31261     13.3  2766 3.2714157               3.46   80     1 0.0074 0.007485937 1.326013 -9.361895
2   31937    31937      0.0     0 1.7116404              -0.14    1     0 0.0074 0.007485937 0.000000 -9.361895
3   33200    31937     19.0  5258 4.1927036               0.85    2     0 0.0074 0.007485937 1.833861 -9.361895
4   32604    31937     38.9  9260 6.2539595              -7.03    3     0 0.0074 0.007485937 2.328253 -9.361895
5   31261    31937     17.6  4158 1.4511050               1.06    4     0 0.0074 0.007485937 1.640549 -9.361895
6   31246    31937     11.3  3333 1.4195458              -2.21    5     0 0.0074 0.007485937 1.466260 -9.361895
7   31285    31937     15.9  4465 2.3303198               0.46    6     0 0.0074 0.007485937 1.698364 -9.361895
8   31297    31937     13.7  3982 0.7931991               1.40    7     0 0.0074 0.007485937 1.605831 -9.361895
9   31218    31937     27.2  6001 1.9189675              -4.05    8     0 0.0074 0.007485937 1.946053 -9.361895
10  31966    31937     45.9 11088 1.8293264              -1.11    9     0 0.0074 0.007485937 2.492213 -9.361895
```

```
glm(event ~ log_dist + offset(logtm),
family = poisson(link = "log"), data = poisson_dat)
```

2.2 Partial likelihood approaches

Consider the second term in Equation (5): their joint product is also known as **partial likelihood** [Cox, 1975].

$$\mathcal{L}^P(\boldsymbol{\beta}) = \prod_{i=1}^{n} \frac{\lambda_{s_i r_i}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta})}{\sum_{(s,r)\in\mathcal{R}_{t_i}} \lambda_{sr}(t_i|\mathcal{W}_{t_i^-};\boldsymbol{\beta})} = \prod_{i=1}^{n} \frac{\exp\left\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i r_i}(t_i)\right\}}{\sum_{(s,r)\in\mathcal{R}_{t_i}} \exp\left\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\right\}} \tag{8}$$

Partial log-likelihood in practice

Log-likelihood :

$$\ell^P(\boldsymbol{\beta}) = \log \mathcal{L}^P(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left[\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i r_i}(t_i) - \log\left(\sum_{(s,r)\in\mathcal{R}_{t_i}} \exp\left\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\right\} \right) \right],$$

Inference technique : **CoxPH regression**;

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \ell^P(\boldsymbol{\beta})$$

Data overview & R code :

```
> head(cox_dat, 10)
   sender receiver dist_min  dist   PT_dist random_intercept_s info event  start         stop log_dist
1   33202    31261     13.3  2766 3.2714157               3.46   80     1 0.0074 0.007485937 1.326013
2   31937    31937      0.0     0 1.7116404              -0.14    1     0 0.0074 0.007485937 0.000000
3   33200    31937     19.0  5258 4.1927036               0.85    2     0 0.0074 0.007485937 1.833861
4   32604    31937     38.9  9260 6.2539595              -7.03    3     0 0.0074 0.007485937 2.328253
5   31261    31937     17.6  4158 1.4511050               1.06    4     0 0.0074 0.007485937 1.640549
6   31246    31937     11.3  3333 1.4195458              -2.21    5     0 0.0074 0.007485937 1.466260
7   31285    31937     15.9  4465 2.3303198               0.46    6     0 0.0074 0.007485937 1.698364
8   31297    31937     13.7  3982 0.7931991               1.40    7     0 0.0074 0.007485937 1.605831
9   31218    31937     27.2  6001 1.9189675              -4.05    8     0 0.0074 0.007485937 1.946053
10  31966    31937     45.9 11088 1.8293264              -1.11    9     0 0.0074 0.007485937 2.492213
```

```
coxph(Surv(start,stop,event) ~ log_dist, data = cox_dat)
```

2.3 Sampled partial likelihood approaches

Nested case-control dataset [Borgan et al., 1995, Vu et al., 2015, Lerner and Lomi, 2020]:

$$D_m = \{(e_i, \mathcal{SR}_{t_i}),\, \mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i},\ |\mathcal{SR}_{t_i}| = m+1,\ i = 1,\ldots,n\}, \tag{9}$$

where:

- $\mathcal{SR}_{t_i}$: **sampled risk set** at t_i, including (s_i, r_i) and m randomly sampled dyads in $\mathcal{R}_{t_i}$.
- m: **number of sampled non-events**.
- Event history and related sampling information are incorporated in a **filtration** $\mathbb{F} = \{\mathcal{F}_t\}_{t\in\mathbb{R}^+}$ where $\mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \le t)$.

Relational event model with sampling:

$$\lambda_{sr,\mathcal{SR}}(t) = \lambda_{sr}(t)\cdot\pi_t(\mathcal{SR}|(s,r)), \qquad \pi_t(\mathcal{SR}|(s,r)) = \frac{m}{|\mathcal{R}_t|-1}\cdot 1_{\{(s,r)\in\mathcal{SR},\mathcal{SR}\subset\mathcal{R}_t\}} \tag{10}$$

Sampled partial likelihood:

$$\mathcal{L}^S(\boldsymbol{\beta}) = \prod_{i=1}^{n} \frac{\lambda_{s_ir_i}(t_i;\mathcal{F}_{t_i^-};\boldsymbol{\beta})\cdot\cancel{\pi_{t_i}(\mathcal{SR}_{t_i}|(s_i,r_i))}}{\sum_{(s,r)\in\mathcal{SR}_{t_i}}\lambda_{sr}(t_i;\mathcal{F}_{t_i^-};\boldsymbol{\beta})\cdot\cancel{\pi_{t_i}(\mathcal{SR}_{t_i}|(s,r))}} = \prod_{i=1}^{n}\frac{\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_ir_i}(t_i)\}}{\sum_{(s,r)\in\mathcal{SR}_{t_i}}\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\}}, \tag{11}$$

that coincides with the likelihood of a **conditional logistic regression** of a stratum of size $m+1$ with the first $k=1$ observations being the cases.

Sampled partial log-likelihood in practice

Log-likelihood :

$$\ell^S(\boldsymbol{\beta}) = \log\mathcal{L}^S(\boldsymbol{\beta}) = \sum_{i=1}^{n}\left[\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_ir_i}(t_i) - \log\left(\sum_{(s,r)\in\mathcal{SR}_{t_i}}\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\}\right)\right]$$

Inference technique : **Conditional logistic regression**;

$$\begin{aligned}\Delta N_{sr}(t_i)|\sum_{s'r'\in\mathcal{SR}_i}\Delta N_{s'r'}(t_i) &= 1 \sim \text{Categorical}\,(\pi_{sr}(t_i))\\ \pi_{sr}(t_i) &= \frac{\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\}}{\sum_{(s',r')\in\mathcal{SR}_{t_i}}\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s'r'}(t_i)\}}\end{aligned}$$

Data overview & R code :

```
> head(sampled_dat, 10)
   sender receiver dist_min  dist   PT_dist random_intercept_s info event  start        stop log_dist
1   33202    31261     13.3  2766 3.2714157               3.46   80     1 0.0074 0.007485937 1.326013
2   31297    31218     15.9  3630 0.7931991               1.40  147     0 0.0074 0.007485937 1.532557
3   31255    31419     19.5  4453 6.7501025              -0.15  318     0 0.0074 0.007485937 1.696166
4   31071    32604     50.3 12994 1.6706971               3.96   54     0 0.0074 0.007485937 2.638629
5   31272    31071     34.6  8183 2.2255955               0.14  277     0 0.0074 0.007485937 2.217354
6   32217    31071    124.3 33190 3.8203727              -3.86  279     0 0.0074 0.007485937 3.531933
7   31324    31262     11.2  2435 1.3001624               6.14  250     1 0.0132 0.013290035 1.234017
8   32217    31297    117.9 44455 3.8203727              -3.86  139     0 0.0132 0.013290035 3.816723
9   31255    32604     49.4 12196 6.7501025              -0.15   58     0 0.0132 0.013290035 2.579914
10  32604    31218     63.9 15186 6.2539595              -7.03  143     0 0.0132 0.013290035 2.784147
```

```
clogit(event ~ log_dist + strata(stop), data = sampled_dat)
```

2.4 Case-1-control partial likelihood approaches

Consider (11) in the particular case of $m = 1$ [Boschi et al., 2025a, Filippi-Mazzola and Wit, 2024].

$$\mathcal{L}^S(\boldsymbol{\beta}) \overset{m=1}{=} \prod_{i=1}^{n} \frac{\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i r_i}(t_i)\}}{\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i r_i}(t_i)\}+\exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i^* r_i^*}(t_i)\}} = \prod_{i=1}^{n} \frac{\exp\{\boldsymbol{\beta}\cdot\Delta\boldsymbol{x}_i\}}{1+\exp\{\boldsymbol{\beta}\cdot\Delta\boldsymbol{x}_i\}} = \prod_{i=1}^{n} \text{logistic}(\boldsymbol{\beta}\cdot\Delta\boldsymbol{x}_i),$$

where:

- (s_i^*, r_i^*): the **sampled dyad** at time t_i, drawn from the risk set $\mathcal{R}_{t_i}$ ($m = 1$);
- $\Delta\boldsymbol{x}_i = \boldsymbol{x}_{s_i r_i}(t_i) - \boldsymbol{x}_{s_i^* r_i^*}(t_i)$: the difference between covariates evaluated for the event and the sampled non-event at time t_i.

Equation (2.4) coincides with the likelihood of a **degenerate logistic regression**:

- without intercept term,
- with constant response equal to 1,
- with covariates equal to the difference between covariates evaluated for the event and the sampled non-event at time t_i.

Case-1-control likelihood in practice

Log-likelihood :

$$\ell^S(\boldsymbol{\beta}) \overset{m=1}{=} \log\mathcal{L}^S(\boldsymbol{\beta}) = \sum_{i=1}^{n}\left[\boldsymbol{\beta}\cdot\boldsymbol{x}_{s_i r_i}(t_i) - \log\left(\sum_{(s,r)\in\{(s_i,r_i),(s_i^* r_i^*)\}} \exp\{\boldsymbol{\beta}\cdot\boldsymbol{x}_{sr}(t_i)\}\right)\right]$$

Inference technique : **Degenerate logistic regression**;

$$Y_i|\Delta\boldsymbol{x}_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\pi_i), \quad \text{logit}(\pi_i) = \boldsymbol{\beta}\cdot\Delta\boldsymbol{x}_i, \quad y_i = 1 \;\; i = 1,\ldots,n; \quad \text{no intercept} \tag{12}$$

Data overview & R code :

```
> head(dat_gam, 10)
   y        time    s1    r1 log_dist1    s2    r2 log_dist2 delta_log_dist
1  1 0.007485937 33202 31261 1.3260134 33200 33202  1.229348    0.096665320
2  1 0.013290035 31324 31262 1.2340169 31966 31255  2.766130   -1.532113487
3  1 0.016123561 31261 33200 0.6333972 31272 31071  2.217354   -1.583956772
4  1 0.016833676 31218 31218 0.0000000 31218 32604  2.821557   -2.821557442
5  1 0.027780092 31285 31246 0.7788661 31297 31071  1.876101   -1.097234562
6  1 0.036779525 31272 31023 2.3787125 31966 32050  3.278728   -0.900015931
7  1 0.045849812 31218 31218 0.0000000 31272 31262  1.149305   -1.149305403
8  1 0.047972385 33202 31966 2.8733951 31255 31937  1.585350    1.288045017
9  1 0.052986957 31272 31419 1.9516082 33200 31313  1.925853    0.025754966
10 1 0.061285646 31071 31023 1.9742199 31071 31246  1.968091    0.006129007
```

```
gam(y ~ -1 + delta_log_dist,
family="binomial"(link = 'logit'), data = dat_gam)
```

3 Non-Linear REMs

Additive relational event model (example): consider Equation (19) with

$$f\left(\boldsymbol{x}_{sr}(t), \boldsymbol{v}_{sr}(t), t\right) = \underbrace{\beta \cdot x^{(1)}_{sr}(t)}_{\text{fixed linear}} + \underbrace{\beta(t) \cdot x^{(2)}_{sr}(t)}_{\text{time-varying}} + \underbrace{f^{(3)}(x^{(3)}_{sr}(t))}_{\text{non-linear}} + \underbrace{\boldsymbol{\gamma} \cdot \boldsymbol{v}_{sr}(t)}_{\text{random}} \tag{13}$$

as a contribution function, including four types of effects:

1. **Fixed linear effect** (FLE): β is time-homogenous and is not a function of $x^{(1)}_{sr}(t)$;
2. **Time-varying effect** (TVE): $\beta(t)$ is a non-linear function of time;
3. **Non-linear effect** (NLE): $f^{(3)}(x^{(3)}_{sr}(t))$ is a non-linear function of $x^{(3)}_{sr}(t)$;
4. **Random effect** (RE): $\boldsymbol{\gamma} = (\gamma_s)_{s\in V}$, γ_s is a Gaussian random variable;

This model can be fitted using a degenerate logistic regression defined in (12).

Time-varying effect. In its simplest form, a **smooth function of time** can be represented as a linear combination of basis functions of **calendar time**.

$$\beta(t) \approx \sum_{j=1}^{q} \theta_j b_j(t), \tag{14}$$

where:

- q: **basis dimension**;
- $\{b_j\}_j$: set of **basis functions**;
- θ_j: **basis coefficients**;

TVE in practice

[Boschi et al., 2025a]
Contribution to the log-event rate :

$$f\left(\boldsymbol{x}_{sr}(t), \boldsymbol{v}_{sr}(t), t\right) = \ldots + \underbrace{\sum_{j=1}^{q} \theta_j b_j(t)}_{\text{TVE}} \times x^{(2)}_{sr}(t) + \ldots$$

Contribution to the case-1-control partial likelihood :

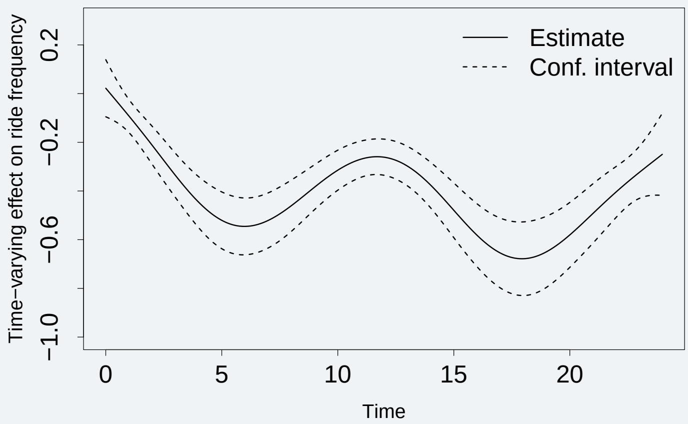
$$\mathcal{L}^S(\boldsymbol{\theta}) \overset{m=1}{=} \prod_{i=1}^{n} \frac{e^{\cdots + \left[\sum_{j=1}^{q} \theta_j b_j(t_i)\right] \Delta x^{(2)}_i + \ldots}}{1 + e^{\cdots + \left[\sum_{j=1}^{q} \theta_j b_j(t_i)\right] \Delta x^{(2)}_i + \ldots}} \;(+\text{ smoothing penalty})$$

Data overview & R code :

```
> head(data)
  y        time    s1    r1  PT_dist1    s2    r2  PT_dist2    PT_dist
1 1 0.001451090 32217 32604 3.820373 32604 31937 6.2539595 -2.4335868
2 1 0.001584378 31937 31285 1.711640 32217 31246 3.8203727 -2.1087323
3 1 0.008190801 31324 31218 1.300162 31272 31261 2.2255955 -0.9254332
4 1 0.011731796 31313 31272 2.346278 31297 32604 0.7931991  1.5530784
5 1 0.014548734 31285 32217 2.330320 31261 31285 1.4511050  0.8792148
6 1 0.018378723 33200 31313 4.192704 31262 31419 1.1147747  3.0779289
```

```
gam(y ∼ -1+ s(time, by = PT_dist), family= "binomial", data = data)
```

Estimated effect (example):



Non-linear effect: In its simplest form, a **smooth function of a covariate** can be represented as a linear combination of basis functions of the **covariate** itself.

$$f(x_{sr}(t)) \approx \sum_{j=1}^{q} \theta_j b_j(x_{sr}(t)) \tag{15}$$

where (as before):

- q: **basis dimension**;
- $\{b_j\}_j$: set of **basis functions**;
- θ_j: **basis coefficients**;

NLE in practice

[Filippi-Mazzola and Wit, 2024]

Contribution to the log-event rate :

$$f\left(\boldsymbol{x}_{sr}(t), \boldsymbol{v}_{sr}(t), t\right) = \ldots \ + \underbrace{\sum_{j=1}^{q} \theta_j b_j(x_{sr}^{(3)}(t))}_{\text{NLE}} + \ \ldots$$

Contribution to the case-1-control partial likelihood :

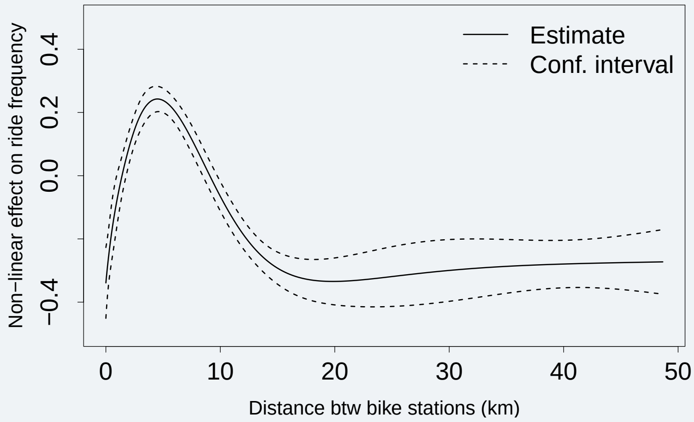
$$\mathcal{L}^S(\boldsymbol{\theta}) \overset{m=1}{=} \prod_{i=1}^{n} \frac{e^{\ldots\ +\ \left[\sum_{j=1}^{q} \theta_j [b_j(x^{(3)}_{s_i r_i}(t_i)) - b_j(x^{(3)}_{s_i^* r_i^*}(t_i))]\right]\ +\ \ldots}}{1 + e^{\ldots\ +\ \left[\sum_{j=1}^{q} \theta_j [b_j(x^{(3)}_{s_i r_i}(t_i)) - b_j(x^{(3)}_{s_i^* r_i^*}(t_i))]\right]\ +\ \ldots}} \text{ (+ smoothing penalty)}$$

Data overview & R code :

```
> head(data)
  y        time    s1    r1  dist1    s2    r2  dist2 ev.i nonev.i
1 1 0.001377150 33202 32604 14.073 32604 31937  9.260    1      -1
2 1 0.001453772 31937 31285  4.404 32217 31246 43.790    1      -1
3 1 0.001979046 33200 31937  5.258 31255 31261  1.881    1      -1
4 1 0.004495738 31246 31937  3.333 31218 31937  6.001    1      -1
5 1 0.007457826 31324 31218  3.970 31272 31261  3.300    1      -1
6 1 0.010580056 31272 31324  4.492 31272 31937  6.689    1      -1
```

```
gam(y ∼ -1+ s(cbind(dist1, dist2),by = cbind(ev.i, nonev.i),
                family= "binomial", data = data)
```

Estimated effect (example):



Random effect: can be efficiently estimated as **0-dimensional splines**, with basis functions taking the value 1 when the level of the random factor is present and 0 otherwise.

$$\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s' = s\} \qquad \boldsymbol{\gamma} = (\gamma_1, ..., \gamma_{|V|})$$

RE in practice

[Boschi et al., 2025a]

Contribution to the log-event rate :

$$f\left(\boldsymbol{x}_{sr}(t), \boldsymbol{v}_{sr}(t), t\right) = \dots \ + \underbrace{\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s = s'\}}_{\text{RE}} + \ \dots$$

Contribution to the case-1-control partial log-likelihood :

$$\ell^S(\boldsymbol{\gamma}) = \log \mathcal{L}^S(\boldsymbol{\gamma}) \overset{m=1}{=} \sum_{i=1}^{n} \log \left(\frac{e^{\dots \ + \gamma_{s_i} - \gamma_{s_i^*} \ + \ \dots}}{1 + e^{\dots \ + \gamma_{s_i} - \gamma_{s_i^*} \ + \ \dots}} \right) - \frac{1}{2\sigma_s^2} \boldsymbol{\gamma} \cdot \boldsymbol{\gamma}$$

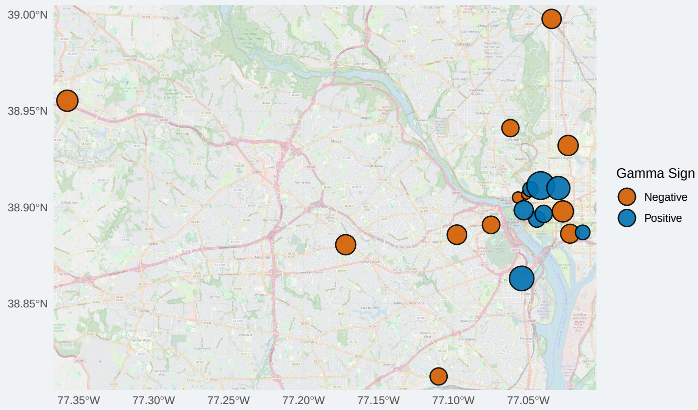
Data overview & R code :

```
> head(data)
     y        time    s1    s2 ev.i nonev.i
[1,] 1 0.003427363 31324 31023    1      -1
[2,] 1 0.041364621 31324 31262    1      -1
[3,] 1 0.042194513 31324 31272    1      -1
[4,] 1 0.049880008 31324 31262    1      -1
[5,] 1 0.051420690 31324 31071    1      -1
[6,] 1 0.051820703 31324 31419    1      -1
```

```
stat_pop <- matrix(factor(c(data$s1, data$s2)), ncol = 2)

gam(y ~ -1+ s(stat_pop, by = cbind(ev.i, nonev.i), bs = "re"),
          method = "REML", family="binomial", data = data)
```

Estimated effect (example):



4 Practical

4.1 Alien species invasions as relational events

a) **Load the input data. Inspect the original data contained in the object `FR`. Have a look at the columns: `LifeForm`, `Taxon`, `Region`, `FirstRecord`, `Source`.**

We firstly show how data arrives from the original source [Seebens et al., 2018]. For each record, we are able to identify:

- the **species** (`Taxon`) of a certain **lifeform** (`LifeForm`)
- that is invading a region (`Region`) in which it was not native and
- that was not seen before the reported **year** (`FirstRecord`).

The Alien Species First Records Database `FR` collects several sources: the source of each FR is reported in the column `Source`.

b) **Interpret this data as a relational event network. Define sender set, receiver set, and time-window.**

We recall that a **first record** is the **year** (t) when a (non-native) **species** s first invades **region** r. Dataset `FR` contains information we need to interpret first records as **relational events** (s, r, t). Particularly, the **sender set** V^S is composed of the unique recorded species, while regions of the world (countries and islands) form the **receiver set** V^R. The **time window** spans from $t_0 = 1880$ to $T = 2005$.

c) **Inspect the object `native`. A species' native range (NR) is the collection of areas where it is indigenous. How would you relate this concept to that of risk set?**

Slightly more liberally, we refer to **native range** as the set of sites where a species was already present before the start of the analysis period, which in this context is 1880. `native` object also contains the FRs, involving insects, that occurred before 1880.The relational event is **non-recurrent**, namely, the event first record, for its definition, occurs only once. This also means that the **risk set** is reducing over time: once the event is observed, it cannot be sampled anymore as a non-event. But also because covariates are endogenous and at each time, the first time included, rely on the knowledge of which first records already occurred and this must be possible at the beginning of the analysis as well.

d) **Read the function `invaded.regions` that allows computing which regions the species invaded before the current year. Which type of covariates does this function allow to compute: exogenous or endogenous, node- or dyad-level, time-dependent?**

This type of function allows us to identify particular instances in the **previous event history** (specifically, past events involving the same species) and to extract **attributes of interest** (e.g., which regions were invaded). This information can then be aggregated into a covariate. Since this covariate is a function of the past network of events, the resulting covariate is **endogenous**. Since we do not store the time of these events - but only the receivers - the covariate will **not be a function of the internal time**.

4.2 Case-1-control dataset

Review the commented function `creating_case_control_dataset`, which utilizes information from the relational event network, noting that the relational event is non-recurrent. Use it to generate the case-control dataset.

To perform inference using the **maximum sampled partial likelihood** with $m = 1$, we construct the **case-control dataset**, which includes, for each first record, the dyad involved in the **observed event** and that involved in the **non-event** randomly selected from the risk-set. It is essential to emphasize that in this scenario, due to the non-recurrent nature of the relational event, the risk set diminishes over time. Once the event occurs, it cannot occur again and thus, it is excluded from the dyads at risk. In order to address the **ties** in the event sequence, we exclude, for each event, all the first records that occurred in the same year from being considered as potential non-events.

The **case-1-control dataset**, with one sampled non-event for each event, represents the starting point of our analysis.

4.3 Covariate computation and model fitting

4.3.1 Linear effect of climatic dissimilarity

a) **Consider the function `climatic_dissimilarity`: given a relational event (and all the relevant information that is required), it computes the minimum temperature difference between the region involved and the previously invaded regions by the species. Apply the function to the events and the sampled non-events and compute the difference. Is it node- or dyad-level? Time-dependent? Exogenous?**

   `data_temperature` contains the names of the countries in the first column and the average values of near-surface air temperature in the second column [Watanabe et al., 2011]. The covariate depends on both the sender and the receiver of the event and can therefore be considered a **dyad-level covariate**. It is time-dependent but does **not incorporate information about internal time**. It is **endogenous**, as it relies on the regions previously invaded by the species, but it also incorporates **exogenous information**.

b) **Estimate the linear fixed effect of climatic dissimilarity and interpret the resulting estimate.**

```
gam_dt.only <- gam(y ~ dt - 1, family="binomial"(link = 'logit'), data=dat.gam)
```

   The coefficient for climatic dissimilarity, which remains **constant over time**, is **negative**. This indicates that the rate of invasions tends to increase if the species has already invaded countries with similar temperatures.

4.3.2 Time-varying effect of trade on species invasions

a) **Consider the function `log_trade`, that, given a relational event (and all the relevant information required), computes the annual trade between the involved region and the previously invaded regions by the species. Apply the function to the events and the sampled non-events, and compute the difference. What kind of covariate is it?**

   In the existing literature, **international trade** has been recognized as a key factor in explaining the spread of alien species. Here, we consider trade as the yearly commerce between already-invaded territories and the involved region. The source information, derived from [Barbieri et al., 2009], which reports trade **flows among countries**, is contained in `data_trade`. Trade

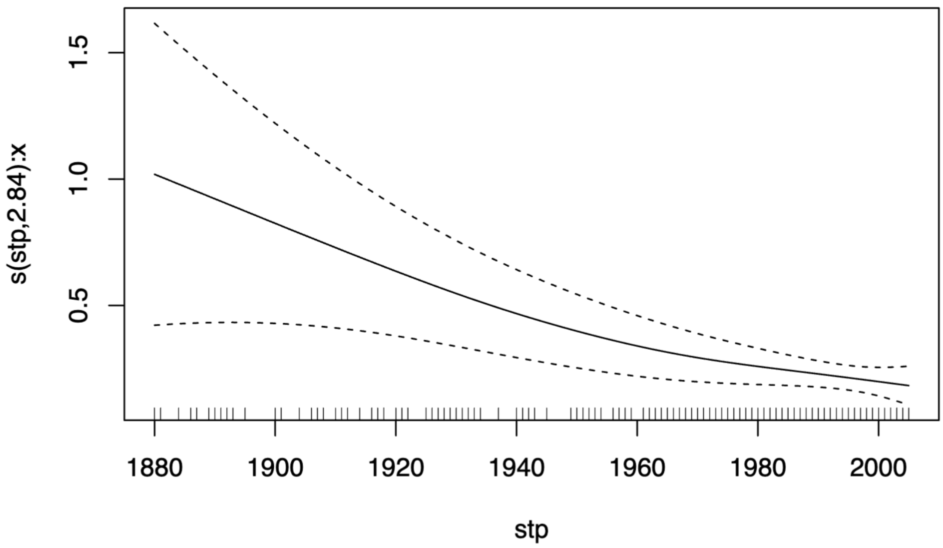


Figure 1: TVE of trade. The effect of trade appears to be decreasing over the time period.

depends both on species and region (and explicitly on time as well). So it is **dyadic** because it involves both the sender and the receiver involved in the relational event. To compute this covariate, we need the **set of invaded regions** by species s by time t and this is an **endogenous** type of information, arising from the network. Nevertheless, **exogenous** information is also present, concerning the flows among countries.

b) **Estimate the time-varying effect of trade and interpret the resulting estimate.**

Let `x.ev` and `x.nv` be $n \times 1$ vectors of trade covariate evaluated for events & non-events. Then: `x = x.ev - x.nv`. Let `stp` be $n \times 1$ vector of event-times. Fit the model incorporating a time-varying effect for `x` and interpret the estimate.

```
gam_tr.only <- gam(y ~ s(stp, by=x) - 1,
                   family="binomial"(link = 'logit'), data=dat.gam)
```

4.3.3 Non-Linear Effect of Distance on Species Invasions

a) **Consider the function `log_distance`, that, given a relational event (and all the relevant information that is required), computes the shortest distance among the regions in which the species is already present at that time. Apply the function to the events and the sampled non-events.** We compute distance for the dyad (species-region) as the shortest distance among the regions in which the species is already present at that time. Distance is computed referring to the closest borders and consequently is equal to zero for neighbours. Source data for this driver consists of the R package geosphere [Hijmans et al., 2017]. `data_distance` is a matrix with rownames and colnames referring to the regions. The corresponding element in the matrix reports the distance between the two.

b) **Estimate the non-linear effect of distance and interpret the resulting estimate.** Let `x.ev` and `x.nv` be $n \times 1$ vectors of distance covariate evaluated for events & non-events and `unit` be $n \times 1$ unit vector.

```
gam_d.only <- gam(y ~ s(X, by=I) - 1,
                  family="binomial"(link = 'logit'), data=dat.gam)
```

4.3.4 Species Invasiveness Random Effect

a) **Create a $n \times 2$ matrix `sp` containing in the first column `spec` levels related to species involved in the events and in the second those involved in the non-events. Is $v_s(t)$ a node- or dyad-level?** It is just reporting the label of the involved species s, so the covariate $v_s(t)$ is monadic.

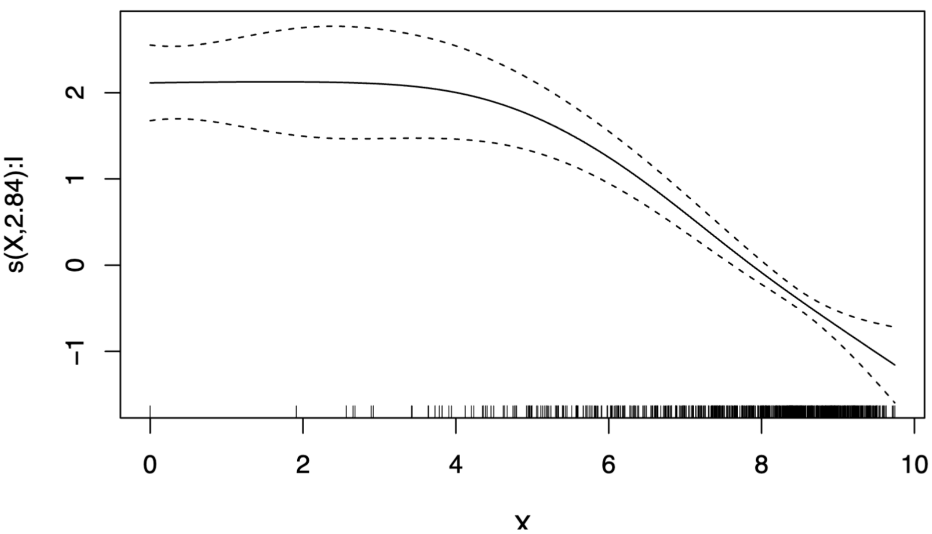


Figure 2: NLE of distance.

```
sp1 <- dat.gam$sp1
sp2 <- dat.gam$sp2
sp <- factor(c(sp1,sp2))
dim(sp) <- c(length(sp1),2)
```

b) **Fit the model incorporating a random effect for species invasiveness and interpret the estimates.**

```
gam_sp.only <- gam(y ~ s(sp, by=I, bs="re") - 1,
                   family="binomial"(link = 'logit'), data=dat.gam)
```

4.4 Model selection

a) **Fit a model with all the components included. Inspect eventual changes in the interpretation and consider the values of the AIC, to understand which model has a better predictive power.**

Among the 5 fitted models, the complete one has better predictive performance. It includes Climatic dissimilarity, Trade, Distance, and the random effect for species invasiveness.

Part II

All you need to know about relational hyperevent modeling

1 Core RHEMs

Relational hyper event sequence: realization of a **marked point process**.

- Sequence of **undirected hyperevents** [Lerner et al., 2025b]:

$$E = \{e_i = (S_i, t_i), i = 1, \dots, n\}, \tag{16}$$

  where:

  - S_i is a subset of the **vertex set** V, composed of the entities involved simultaneously in the i-th relational hyperevent, i.e. the **participants**;
  - t_i: **time** of the i-th relational hyperevent;

  E.g. **Les misérables**, whose chapters can be considered as hyperevents where actors co-appear.

$$E = \{(\{\text{MY, NP, MB}\}, t_1), (\{\text{MY, ME, MB}\}, t_2), (\{\text{MY}\}, t_3), (\{\text{MY, ME, CL, GE, MC, MB}\}, t_4), (\{\text{MY, ME, MB}\}, t_5), \dots\}$$

- Sequence of **directed hyperevents** [Lerner et al., 2025a, Boschi et al., 2025b]:

$$E = \{e_i = (S_i, R_i, t_i), i = 1, \dots, n\}, \tag{17}$$

  - S_i is a subset of the **sender set** V^S, representing the group of senders of the i-th relational hyperevent;
  - R_i is a subset of the **receiver set** V^R, representing the group of receivers of the i-th relational hyperevent;
  - t_i: **time** of the i-th relational hyperevent;

  E.g. **Scientific innovation**, where each paper publication is a hyperevent (a group of authors publishes a paper citing a set of papers).

$$E = \{(\{2,3\}, \{A,B\}, t_1 = 1950), (\{1,2,4\}, \{A,C\}, t_2 = 1951), (\{4\}, \{A,C\}, t_3 = 1953)\}$$

Remark: an undirected hyperevent can be represented as a directed hyperevent with an empty receiver set, i.e. $(S_i, t_i) = (S_i, R_i = \emptyset, t_i)$ is also referred to as a **meeting**.

In both cases, n is the **number of hyperevents** and i the **index of a generic hyperevent**.

Longitudinal network / multivariate counting process [Perry and Wolfe, 2013]:

$$\{N_{SR}(t)\}_{S \subseteq V^S, R \subseteq V^R, t \geq 0}, \tag{18}$$

where:

- $N_{SR}(t)$: number of interactions from S to R until time t.
- V^S: **set of senders** in the system;
- V^R: **set of receivers** in the system;

If senders/receivers belong to the same group, then $V^S = V^R = V$ (**vertex set**), with $p = |V|$.

Doob-Meyer decomposition theorem:

$$N_{SR}(t) = \lambda_{SR}(t) + M_{SR}(t) = \int_0^t \lambda_{SR}(u)\,\mathrm{d}u + M_{SR}(t)$$

where:

- $\Lambda_{SR}(t)$: **cumulative intensity process** (structural part);
- $M_{SR}(t)$: **Martingale residual process** (noisy part);
- $\lambda_{SR}(t)$: **hazard** (also: **rate**, **intensity**): instantaneous rate at which (S, R) occurs at t.

Relational hyper event model (RHEM) [Lerner et al., 2021, Lerner and Lomi, 2023]:

$$\lambda_{SR}(t) = W_{SR}(t) \times \lambda_0(t) \times \exp\left\{ f\left(\boldsymbol{x}_{SR}(t), t\right) \right\} \tag{19}$$

where:

- $W_{SR}(t)$: **risk indicator** – it is equal to 1 if (S, R) is at risk at time t;
- $\lambda_0(t)$: **baseline hazard** (including global risk determinants);
- $f(\boldsymbol{x}_{SR}(t), t)$: **event log-rate contribution**. E.g. **Linear contribution** $f(x_{SR}(t), t) = \beta \cdot x_{SR}(t)$
  - $f(\cdot, t)$: **contribution function** (also, **effect**);
  - $\boldsymbol{x}_{SR}(t)$: edge-specific **covariates**;
  - t: **calendar time**.

(S, R) satisfying $W_{SR}(t) = 1$ compose the **risk set** $\mathcal{R}_t$.

1.1 Exogenous & endogenous drivers for undirected hyperevents

Exogenous and endogenous drivers are measured by means of **covariates** $x_S(t)$. Covariates can be seen as aggregations or summaries of information derived from **attributes** $z_s(t)$ of nodes in subsets $S' \subseteq S$.

- **Exogenous covariates**: do not depend on the network of (previous) events. One can consider both the **main effect** of an exogenous feature and the effect of related **homophily**.

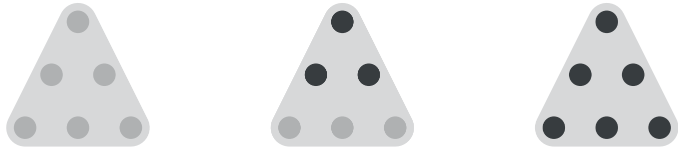
  E.g. **Gender**. Gender is a binary[1] feature that can be evaluated at each node. The covariate capturing the main effect can be computed as the average of the feature over the nodes in the set S.

  $$\text{avg.z}(S, t) = \sum_{s \in S} z_s(t)/|S|$$

  Gender **dissimilarity** can be captured using the following covariate. A negative effect of this covariate indicates the presence of **homophily**.

  $$\text{diff.z}(S, t) = \sum_{\{s, s'\} \in \binom{S}{2}} \frac{|z_s(t) - z_{s'}(t)|}{\binom{|S|}{2}}$$

[1] only for mathematical purposes

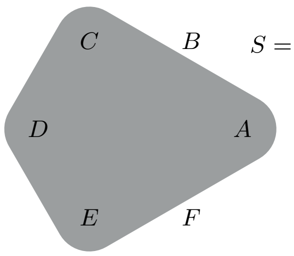


$$\text{avg.z}(S_1) = 0 \qquad \text{avg.z}(S_2) = 0.5 \qquad \text{avg.z}(S_3) = 1$$
$$\text{diff.z}(S_1) = 0 \qquad \text{diff.z}(S_2) = \tfrac{9}{15} \qquad \text{diff.z}(S_3) = 0$$

- **Endogenous covariates**: depend on the network of (previous) events and allow us to measure **relational mechanisms**.

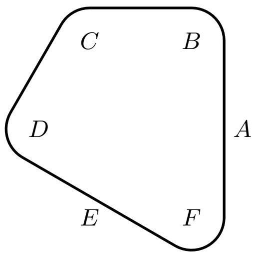
  E.g. **Subset repetition**: the average number of prior joint events over all subsets $S' \subseteq S$ of size ρ. As before, we evaluate the attribute (in this case, the **formation of a hyperedge** in the past) for all possible subsets and then **aggregate** the results to obtain the covariate.

$$\text{hy.deg}(S', t) = \sum_{e_i : t_i < t} \mathbf{1}(S' \subseteq S_i)$$
$$\text{sub.rep}^{(\rho)}(S, t) = \sum_{S' \in \binom{S}{\rho}} \frac{\text{hy.deg}(S', t)}{\binom{|S|}{\rho}}$$



$S = \{B, C, D, F\}$

$$\text{sub.rep}^{(1)}(S) = 2/4$$
$$\text{sub.rep}^{(2)}(S) = 1/6$$
$$\text{sub.rep}^{(3)}(S) = 0/4$$



2 Practical (1) with `eventnet`: Les Misérables

2.1 Introduction to `eventnet`

`eventnet` (`https://github.com/juergenlerner/eventnet`): software for the statistical analysis of **networks of relational (hyper) events**.

To use `eventnet`, **download the file** eventnet-1.3.jar and start the program by double-clicking on the JAR file, which opens the eventnet graphical user interface (GUI).

`eventnet` is written in Java and needs the Java runtime environment (JRE), Java 8 (JRE version 1.8) or higher. You can check the current version of Java installed on your computer by typing the command

```
java -version
```

You can find a copy of the `eventnet-1.3.jar` file in the folder:

```
REM-tutorial/02-All-you-need-to-know-RHEM/01-Les-Miserables
```

2.2 Les Misérables: Data and Covariate Computation

This tutorial is inspired by the analysis conducted in Lerner et al. [2025b], where, for illustrative purposes, RHEM is applied to the **network of actor co-appearances in *Les Misérables***, compiled by Donald Knuth [Knuth, 1993]. The data comprises 80 actors (co-)appearing in one or several of 288 chapters. The (binary[2]) gender of the actors is known.

a) **Given the information above, interpret this data as a relational hyperevent network. Define the components of the hyperevent sequence, the vertex set, its cardinality, and the available attributes.**

[2] only for mathematical purposes

- **80 actors** $|V| = 80$ (co-)appearing in one or several of **288 chapters** ($n = 288$);
- $e_i = (S_i, t_i)$ is a hyperevent (a chapter) involving a set of co-appearing actors S_i. In this case, t_i is simply a counter for the chapter;
- female(i) is a binary gender[3] attribute for each actor. 35% of the actors are female.

b) **Open the file `jean_events.csv`. How are the events reported? Does it contain only event information?**

The file `jean_events.csv` contains three types of entries (which can be distinguished using the column `type`), namely `add.actor`, `is.female`, and `chapter`. Each row involves only one actor, but hyperevents (chapters) are reconstructed by grouping rows with the same chapter identifier.

One particular note concerns the column `add.actor`. This column provides information about which actors could potentially participate in an event at a given point in time. As such, it allows us to construct the risk set at each time point, that is, the set of actors that could have constituted the participant list of an event (but did not).

c) **Open the configuration file `jean.config.small.txt` in `eventnet` and inspect the different windows.**

```
file -> open configuration
```

To answer this question, we explicitly refer to the tutorials Basic tutorial and RHEM first steps.

The `eventnet` configuration informs the software about how the input data should be interpreted and processed.

We can explore the different windows:

(i) files : allows you to specify one or more input files, define the corresponding CSV settings (e.g., delimiter, quote character, header, etc.), and set an output directory. In the given example, there are three columns named `event.id`, `participant`, and `type`. The Files tab for this application is shown in Figure 3.

(ii) events : allows you to map different components of an event to columns of the input file and define whether the event network is one-mode or multi-mode.

Event components:

(a) `SOURCE`: identifies who initiates the event;

(b) `TARGET`: identifies who receives the event. Undirected hyperevents should map `SOURCE` and `TARGET` to the same column indicating the participants;

(c) `TIME`: the time of the event. Time can be expressed in several ways, including integers, decimal numbers, or date–time strings. Additional information must be specified in the Time tab. If time is implied, events are implicitly numbered 1, 2, 3, ..., and these numbers are taken as the event time;

(d) `TYPE`: information about what the source does to the target. If the type is implied, it is implicitly set to `EVENT`;

(e) `WEIGHT`: specifies the strength, magnitude, or intensity of the event. If weight is implied, it is implicitly set to one;

(f) `EVENT_INTERVAL_ID`: marks sub-sequences of simultaneous events. It must necessarily be mapped to a column in the input file;

(g) `NETWORK_ID`: marks sub-sequences that occur within the same event network.

[3] only for mathematical purposes

In the `network mode` section, you can specify whether the network is one- or multi-mode. In our case, the network is one-mode and includes actors only. Once the column for `TYPE` has been set, you can click on the button `learn event types from file`. It is then possible to view the different event types in the section `assign event type to node sets`. The Events tab for this application is shown in Figure 4.

(iii) time : in this window, we focus our attention on the `event interval type` section, which defines which events are considered simultaneous. Several options are available: each event can be treated as occurring alone (`EVENT`); events with the same time value can be considered simultaneous (`TIME`); events occurring within the same time unit can be treated as simultaneous; and finally, our case: events that are assigned the same event interval ID are considered simultaneous (`EVENT_INTERVAL`). Whenever richer information about time is available, it is possible to specify the format and the decimal precision. The Time tab for this application is shown in Figure 5.

(iv) attributes : Attributes include information about past events and assign values to nodes, to the entire network, or to hyperedges of any size. In this example, we define two `NODE_LEVEL`, one `DYAD_LEVEL`, and one `UNDIR_HYPER_LEVEL` attribute. The values of attributes generally change over time. Attributes are then used to define explanatory variables (statistics).

`NODE_LEVEL` :

- **at.risk**: a function of events of type `add_actor`.

  The type is `DEFAULT_NODE_LEVEL_ATTRIBUTE`. The update type is `SET_VALUE_TO`, meaning that the values of this attribute are set to (rather than incremented by) the values of the events specified in the `updates by events` section at the bottom. The direction set to `OUT` means that the source node of the event has its value updated. In this case, using `IN` would lead to the same result.

- **is.female**: defined in the same way as **at.risk**.

`UNDIR_HYPER_LEVEL` :

- **prior.events**: a function of events of type `chapter`.

  The type is `DEFAULT_UHE_ATTRIBUTE`. For each hyperevent and time-point, the attribute counts the number of past events and whose set of participants are exactly the members of the hyperevent. The update type is `INCREMENT_VALUE_BY`, meaning that the values of the events specified in the `updates by events` get added to the current attribute value on the same hyperedge.

`DYAD_LEVEL` :

- **prior.events.dyadics**: a function of events of type `chapter`. This allows to compute, for each pair of actors, the number of previous events in which both jointly participated.

  The type is `DYAD_LEVEL_ATTRIBUTE_FROM_UHE`.

The Attributes tab for this application is shown in Figure 6.

(v) statistics : Hyperedge statistics are computed by aggregating attributes and, therefore, might be functions of previous history of events. The values of hyperedge statistics are computed for all observed hyperevents and for all, or a subset of, non-events. Even though the configuration file includes information about closure, we focus on five statistics in this tutorial:

(a) **num.actors**: a statistic of type `UHE_SIZE_STAT`. It computes the size of the hyperevent and requires no additional settings.

(b) **individual.activity** and **dyadic.activity**: both are of type `UHE_SUB_REPETITION_STAT`. According to the discussion on subset repetition, these statistics rely on the attribute prior_events. The individual and dyadic versions differ in the hyperedge size, which is equal to 1 and 2, respectively. At the bottom, one can specify how to aggregate the values over the different participant subsets; in this case, they are summed (SUM).

(c) **female** and **diff.female**: both are of type UHE_NODE_STAT. They are computed from the attribute is.female and aggregated by summing (SUM) to evaluate the main effect, and by SUMABSIDFF to evaluate homophily. Figure 7 reports the preview of the Statistics tab.

(vi) observations : this tab allows you to specify which hyperevents are considered and which ones, and how many, can be sampled as non-events. In the uploaded configuration, the observation is named EVENTS_CONDITIONAL_SIZE and has type COND_SIZE_UHE_OBS. It considers only events of type event (and not the other event type add_actor) and - importantly - has the option apply case-control sampling checked. The observation compares each observed hyperevent with all hyperevents of the same size. The Observations tab is reported in Figure 8.

d) **Load the input data** jean_events_EVENTS.csv **located in** 01-Data**. Is this a case-1-control dataset? If not, how do you think it can be constructed?**

The object 'data' is not a case-1-control dataset. To construct it, we first create two separate objects for the datasets containing events and non-events, respectively. We then merge these datasets using a set of variables that identify which event each non-event refers to. Among the non-events satisfying this condition, we sample only one.

2.3 Model fitting

a) **Based on the data structures you were given and those you constructed, which inference techniques can you apply?**

Option 1. Since we do not have access to timing information for the events and we have a case-control sampling with 20 non-events, we use the clogit function in R to fit our model (conditional logistic regression). When applied in this context, clogit internally calls the coxph routine. We find a **positive effect of individual and dyadic activity**. This suggests that prior (co-)presence in previous chapters - whether alone or in pairs - increases the rate of participating in events. A negative effect for diff.female suggests a positive effect for gender homophily, favouring co-appearance with other actors of the same gender.

Option 2. Using the case-1-control dataset, it is also possible to fit a degenerate logistic regression without an intercept, where the covariates are defined as the differences between the covariates for the events and those for the non-events, and the response variable is constant and equal to 1. As before, we find a positive effect of individual, dyadic and a negative main effect for female gender and a positive effect for gender homophily.

3 Non-linear RHEMs

3.1 Directed hyperevents

This section reviews some examples of **directed hyperevents**, specifically referring to the paper publication hyperevent defined above. Data concerning directed hyperevents manifest in the form of Equation (17).

3.2 Exogenous & endogenous drivers for directed hyperevents

Endogenous covariates: depend on the network of (previous) events and allow us to measure **relational mechanisms**.

E.g. **Subset repetition**. The definition can be extended in the presence of directed hyperevents by considering the average number of prior joint events involving a subset of senders and a subset of receivers of a given size in the past event history.

$$
\begin{aligned}
\mathsf{cite}^{\mathsf{aut-pap}}(S,R,t) &= \sum_{t_i<t} \omega(t-t_i)\cdot 1_{\{S\subseteq S_i \cap R\subseteq R_i\}} \\
\mathsf{subrep}^{(\rho,\ell)}(S,R,t) &= \sum_{(S',R')\in\binom{S}{\rho}\times\binom{R}{\ell}} \frac{\mathsf{cite}^{\mathsf{aut-pap}}(S',R',t)}{\binom{|S|}{\rho}\times\binom{|R|}{\ell}}
\end{aligned}
\tag{20}
$$

We propose defining subset repetition as an aggregation of the author–paper citation attribute (see Equation (25) for more information). However, subset repetition is a very general concept and can be extended well beyond the notion of citation.

3.3 Beyond Linearity

What can break Linearity and Time-Homogeneity?

Saturation : individuals face cognitive overload and become unable to effectively process additional information.

Forgetfulness : propensity to lose track of past details over time.

External Changes : shifts in the environment, technology, or social norms that alter how factors influence the system.

We adapt the notation introduced in Equations (14) and (15) to distinguish the indices of the basis functions when combining the two types of functions.

1. **Fixed linear effect**:

$$
f^{\mathsf{FLE}}(x_{SR}(t),t) = \beta\cdot x_{SR}(t) \tag{21}
$$

2. **Time-varying effect**:

$$
f^{\mathsf{TVE}}(x_{SR}(t),t) = \alpha(t)\cdot x_{SR}(t), \qquad \alpha(t) = \sum_{l=1}^{L} \alpha_l\, a_l(t) \tag{22}
$$

3. **Non-linear effect**:

$$
f^{\mathsf{NLE}}(x_{SR}(t)) = \sum_{q=1}^{Q} \beta_q\, b_q(x_{SR}(t)) \tag{23}
$$

Joint time-varying non-linear effect: coefficients of basis evaluations of covariates are not fixed but vary over time.

$$
f^{\mathsf{TVNLE}}(x_{SR}(t),t) = \sum_{q=1}^{Q}\left(\sum_{l=1}^{L}\alpha_{ql}a_l(t)\right) b_q\left(x_{SR}(t)\right) \tag{24}
$$

To improve the **model fitting stability** when using these types of effects it is possible to:

1. Apply a **monotonic transformation** to each covariate x:

$$
\dot{x} = 1 - \exp\left(-\frac{x}{\epsilon}\right)
$$

   - Choose ϵ so that the transformed covariate (evaluated for events) is as close as possible to a uniform distribution.
   - Apply the same transformation to both events and non-events.

2. Transform event times t using the **empirical cumulative distribution function**:

$$
\dot{t} = \mathrm{ecdf}_t(t)
$$

4 Practical (2): Scientific Innovation

4.1 Aminer Citation Network

a) **Load the input data** `dat_gam.RData` **located in** `01-Data`**. Is this a case-1-control dataset?**

We begin the second part of the tutorial by uploading a new dataset. This dataset has been sampled from an existing dataset used in [Lerner et al., 2025a, Boschi et al., 2025b].

The original dataset consists of the DBLP Citation Network V14, restricted to journal articles, and is derived from the Aminer Citation Network [Tang et al., 2008].

Code to pre-process and filter the data is publicly available at:

https://github.com/juergenlerner/eventnet. To allow you to upload the data and experiment in real time, we provide a sampled version of the original dataset, consisting of $n = 30\,136$ rows.

b) **Interpret this data as a relational hyper event network. Define the relational hyperevent unit, the sender set, the receiver set, and the time-window.**

As discussed in the theoretical part of the workshop, a relational hyperevent is defined as the formal **publication of a scientific work**, marking the moment when a group of authors presents their research together with a set of citations listed in the reference section.

$$\{r_i = (t_i, S_i, R_i), i = 1, \ldots, 10\,046\}$$

Here, r_i represents the **journal article** published at time t_i by a **group of authors** in $S_i \subseteq V^S_{t_i}$ **citing journal articles** in $R_i \subseteq V^R_{t_i}$.

These events can be represented as time-stamped directed hyperedges of a two-mode network. There are two types of nodes: authors (**sender set**) and cited papers (**receiver set**). When a new hyperedge occurs at time t_i, a new publication r_i – authored by a set of authors in S_i and citing a set of papers in R_i – enters the system and can be cited by other papers at later time points.

This two-mode network representation also makes it possible to examine more complex relationships, such as author-cites-paper, paper-cites-paper, and author-cites-author. These three relations are part of a broader class that we will employ later.

The sampled events span **from 1939 to 2023**. The provided dataset is already in a case–1–control formulation, which is explored in more detail in the following section of the tutorial.

4.2 Case-1-control dataset

a) **Explore the case-1-control dataset by making use of observation #481 as an example.**

This data set contains 84 variables (quite a large number!). As in the previous tutorials, we provide you with the case–control data set already computed, but we also try to explain the rationale behind its construction.

Fortunately, we do not need to interpret all 77 variables but just around half of them :). Why is that? For each row, the same piece of information is evaluated twice: once for the event and once for the non-event.

We (probably) understand by now that the statistical unit of relational event modeling is the triple composed of the sender, the receiver, and the time of the event. Where can we find this information? In the second, third, and fourth columns, respectively, named `SOURCE_ev`, `TARGET_ev`, and `TIME_ev`. As we did in the other applications, one non-event is sampled for each event, and the same basic information is stored for the non-event.

4.3 Attribute and covariate computation

a) **Inspect the attributed employed in this tutorial: Author–paper citations and Out-degree.**

We will not dive into the details of `eventnet` covariate computation here (if you are curious, you can find both a tutorial and a configuration file at https://github.com/juergenlerner/eventnet).

However, it is important to understand that, in this application, building the covariates - computed as aggregations of attributes - may require several basic blocks.

We focus on a subset of the hyperedge covariates studied in Lerner et al. [2025a], Boschi et al. [2025b]. Following those works, we incorporate a weight factor ω to model the temporal decay of past events.

The necessary basic blocks (for this tutorial) are the following:

(a) **Author–paper citations**: the extent to which a set of authors has cited a set of papers in joint publications.

$$\text{cite}^{\text{aut-pap}}(S,R,t) = \sum_{t_m<t} \omega(t-t_m)\cdot 1_{\{S\subseteq S_m \cap R\subseteq R_m\}} \tag{25}$$

(b) **Out-degree**: the length of the reference list of a paper.

$$\text{out\_degree}(r,t) = \sum_{t_m<t} \omega(t-t_m)\cdot 1_{\{r=r_m\}}\cdot |R_m| \tag{26}$$

where, for all of them,

$$\omega(t-t_m) = \exp\Big\{-(t-t_m)\frac{\log 2}{T_{\frac{1}{2}}}\Big\},$$

and $T_{\frac{1}{2}} > 0$ is the half-life period, set to three years.

Additional information on further attributes that can be computed for this dataset can be found in Lerner and Lomi [2023], Boschi et al. [2025b].

b) **Inspect the covariates that are present and the data and focus on: Prior papers, Difference in prior papers, Paper outdegree popularity, Paper citation popularity.**

The provided dataset contains the all the covariates employed in Lerner et al. [2025a] and Boschi et al. [2025b]. In this tutorial, we focus on 4 of them: **Prior papers**, **Difference in prior papers**, **Paper outdegree popularity**, **Paper citation popularity**.

Several of the following covariates are defined as subset-repetition, defined in (20).

(a) **Prior papers**: for a set of authors S, the average number of prior papers (downweighted by the elapsed time). The covariates is a measure of past publication activity of the authors in the sender set.

$$\text{prior\_papers}(t,S,R) = \text{subrep}^{(1,0)}(S,R,t) \tag{27}$$

(b) **Difference in prior papers**: for a set of authors S, it captures the heterogeneity of the past publication activity. A positive parameter would suggest a large dispersion in the number of prior publications.

$$\text{difference\_in\_prior\_papers}(t,S,R) = \sum_{\{s,s'\}\in\binom{I}{2}} \frac{|\text{cite}^{\text{aut-pap}}(\{s\},\emptyset,t) - \text{cite}^{\text{aut-pap}}(\{s'\},\emptyset,t)|}{\binom{|S|}{2}} \tag{28}$$

(c) **Paper outdegree popularity**: average length of reference lists of a set of papers R.

$$\text{paper\_outdegree\_popularity}(S, R, t) = \sum_{r \in R} \frac{\text{out\_degree}(r, t)}{|R|} \tag{29}$$

(d) **Paper citation popularity**: for a set of papers R, average number of past citations (down-weighted by the elapsed time). The covariates is a measure of past citation popularity. A positive parameter would be in lign with the idea of the "rich gets richer".

$$\text{paper\_citation\_popularity}(S, R, t) = \text{subrep}^{(0,1)}(S, R, t) \tag{30}$$

Additional information on further covoriates that can be computed for this dataset can be found in Lerner and Lomi [2023], Boschi et al. [2025b].

4.4 Model fitting

In Lerner and Lomi [2023], the results for the linear model fitted on the complete dataset, including all available covariates, are reported. Similarly, in Boschi et al. [2025b], the results for the non-linear models—also based on the complete dataset—are presented.

4.4.1 Linear RHEM

a) **Fit a linear model with Prior papers, Difference in prior papers, Paper outdegree popularity, and Paper citation popularity. Is it more likely to publish with an author who has a similar number of prior papers to yours?**

The dataset does not include differences of the covariates in their untransformed form. These differences are computed for all variables, even though only a subset of them is used in practice.

By fitting a linear model, we find a **positive value** for the covariate **Difference in prior papers**, which indicates a negative homophily effect in prior publication activity among the actors involved in an event. In general, this result suggests that teams of authors are typically composed of authors with large differences in the number of papers they have previously published, for example, a PhD student together with their supervisor.

4.4.2 Beyond linearity

a) **Fit a linear model with Prior papers, Difference in prior papers, Paper outdegree popularity, and Paper citation popularity. Let the effect of the latter be either time-varying, non-linear, or jointly time-varying and non-linear. Has the tendency to publish with an author who has a similar number of prior papers changed over time? Does it exhibit saturation? Both?**

An implicit assumption when fitting any regression model using maximum likelihood estimation is that the covariate processes have finite second moments. Practically speaking, this means that, for improved model-fitting stability, the covariates should not exhibit major outliers.

Lerner and Lomi [2023] address this issue by applying a square-root transformation to the non-negative covariates. Boschi et al. [2025b], instead, map each covariate - as well as the time variable - from its original scale into the interval $[0, 1]$, in a way that distributes the values as uniformly as possible. In this dataset, you will find the covariates transformed according to the second approach.

The plot of the **time-varying effect** shows an increase; the confidence bands are very large at the beginning, probably due to the nature of the data (largest part of the data are from recent years). The effect is significantly time-varying, however. It is also possible to see a slight decrease in the effect at the end of the time window, where, instead, confidence intervals are quite narrow.

The plot of the **non-linear effect** clearly shows a non-monotonic pattern. Apparently, authors tend to publish together when they have very similar levels of experience; otherwise, the tendency to collaborate drops immediately. This tendency then increases again as the difference grows, as in the classic example of a PhD student and their professor. The effect appears to reach saturation and eventually decreases for very large values of the covariate. This seems to highlight an "optimal" level of variation in prior publications that maximizes this covariate's contribution to the log-rate of coauthoring a new paper.

By allowing **both effects simultaneously**, we can see that the non-linear effect remains clearly present, while the decrease/increase over time is visible only at the very beginning of the time window. By comparing the AIC values of the models, we find that jointly combining the effects is not worthwhile, whereas a model with only a non-linear effect is preferred.

Of course, all these conclusions are based on a limited dataset and on a reduced set of covariates.

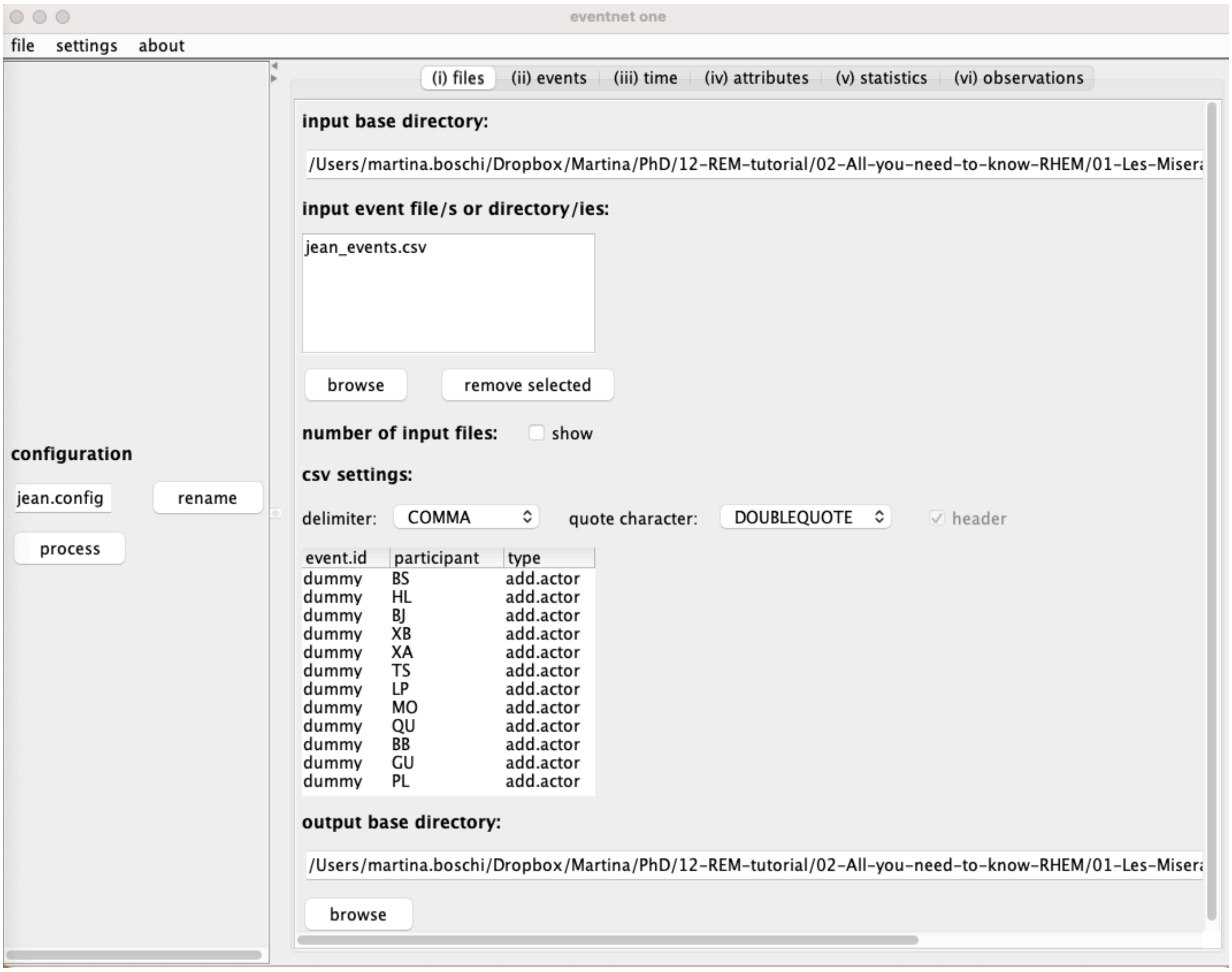


Figure 3: Files tab where you can specify input directory, CSV settings, and the output directory.

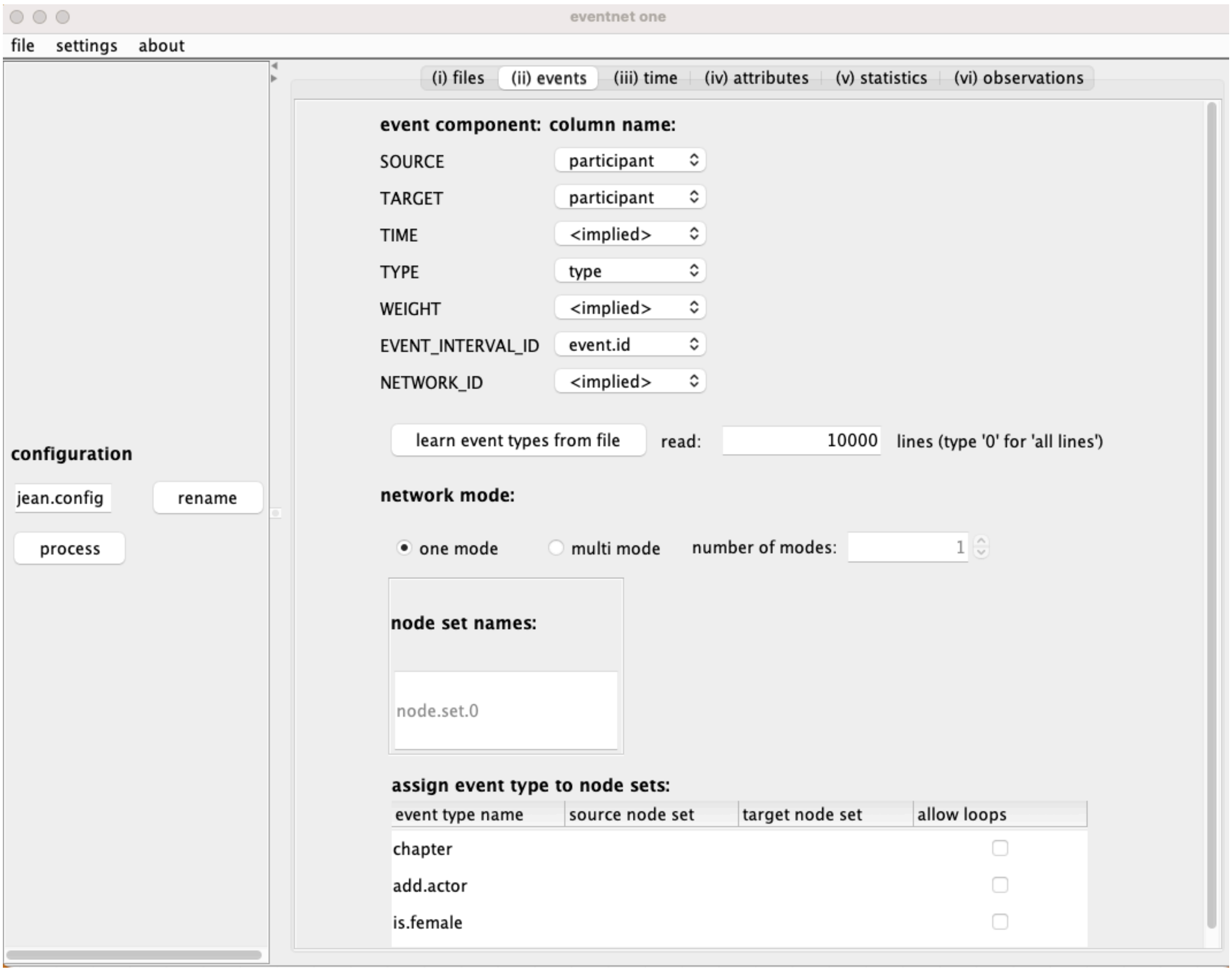


Figure 4: Events tab where you can specify which event component can be found in which column of the CSV file.

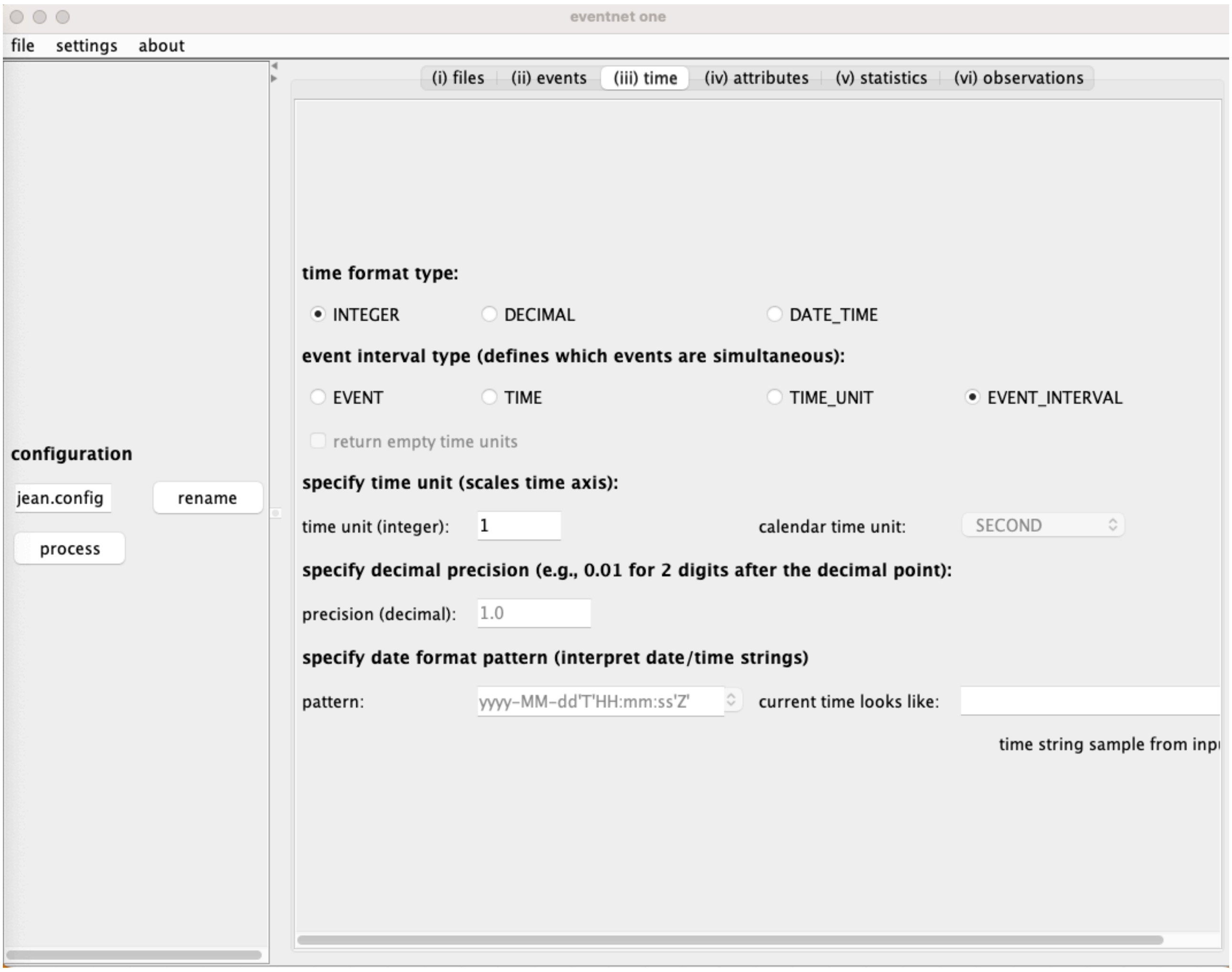


Figure 5: Time tab where you can specify the format of the time variable and which events are considered as simultaneous.

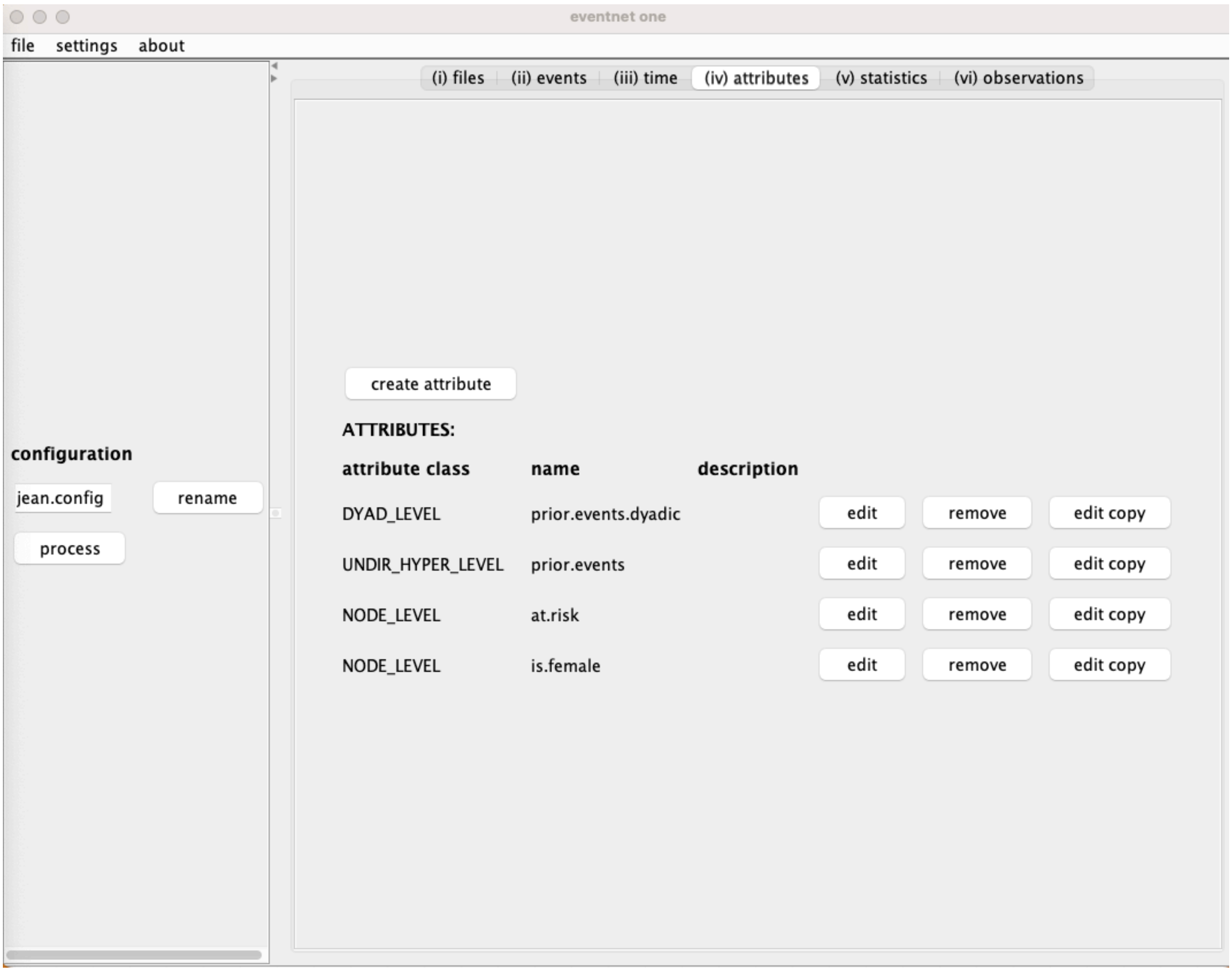


Figure 6: Attribute tab allows to specify and compute attributes for nodes, dyads, entire network, or hyperedges.

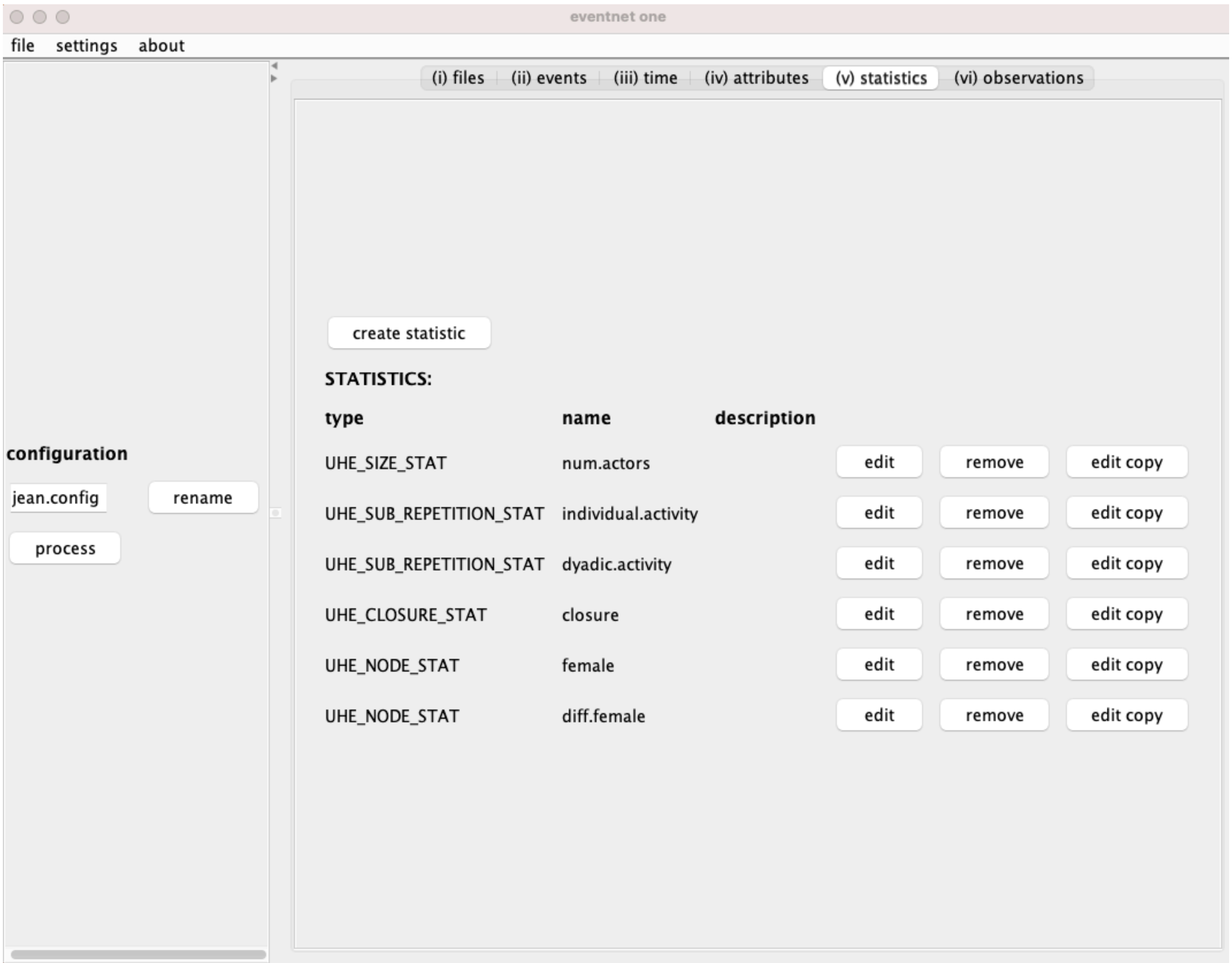


Figure 7: Statistics are the variables that explain the event rate as a function of previous history of events, that is, as a function of attributes defined in the attribute tab.

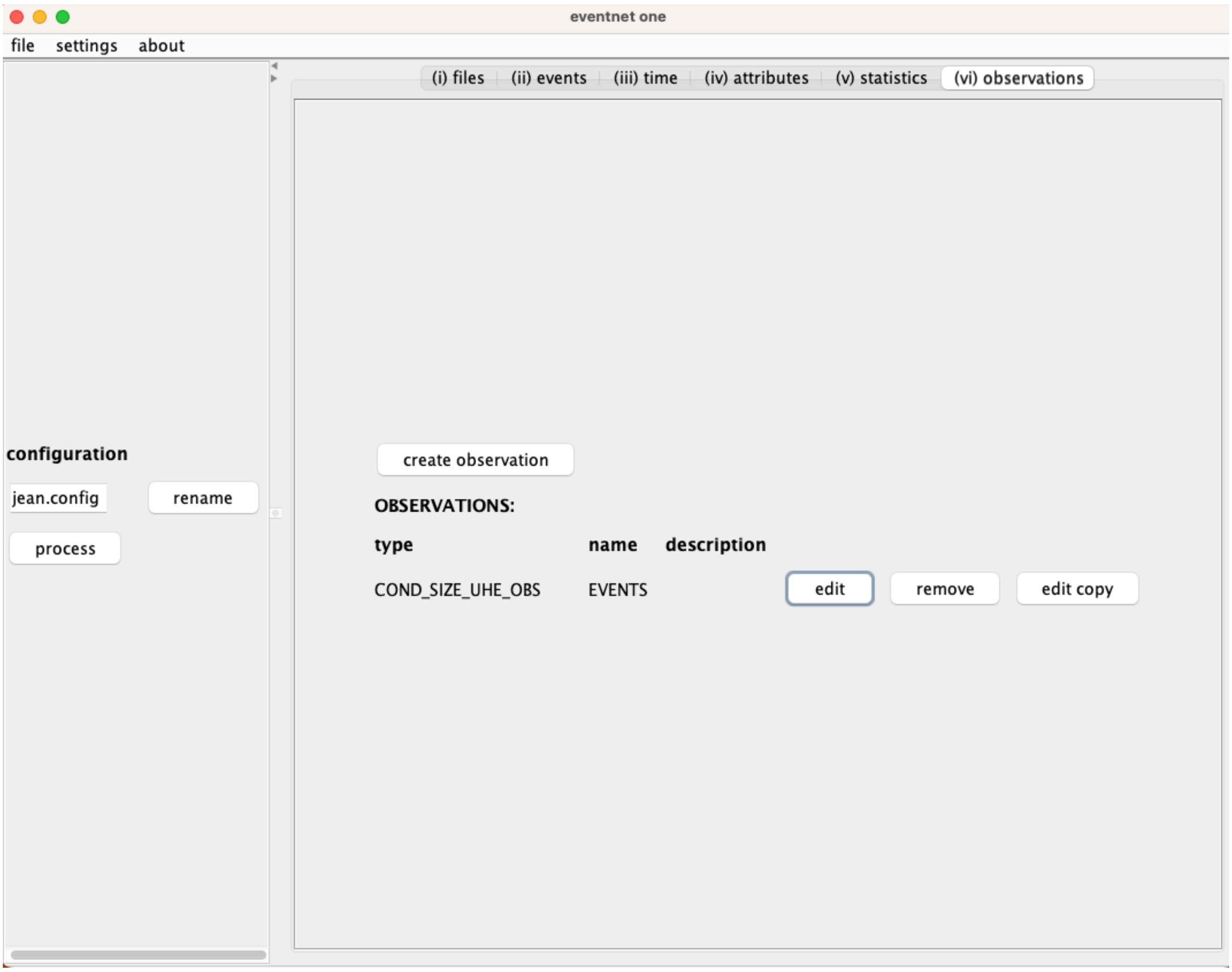


Figure 8: Observation generators (observations) allow the definition of the risk set and how to sample from it.

Index

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